

Simon Chen

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EDUCATION

Boston University

Bachelor of Arts in Computer Science

Boston, MA

Graduated: Spring 2026

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Distributed Systems, Databases.

PROJECTS

Persephone | *Python, Cython/C++17, Rust, Modal, FAISS*

May 2026

- Built and operated a production Kalshi BTC prediction-market trading system that generated \$8K net PnL and traded \$200K volume in its first 4 live days, owning market-data ingestion, theo serving, strategy evaluation, exchange execution, risk gates, and operator controls.
- Engineered a GBM theo layer to replace KDE inference, reducing model query latency from $\sim 1.5\text{ms}$ to $\sim 20\mu\text{s}$ per query ($\sim 75\times$), while preserving the live theo interface used by strategy evaluation, cache publication, and operator tooling.
- Reduced backtest and parameter-sweep iteration from ~ 14 hours on local MacBook CPUs to ~ 15 minutes on Modal NVIDIA GPUs by moving diagnostics, tick backtests, and sweeps onto cloud workers with FAISS GPU search and CuPy-vectorized math.
- Optimized KDE serving with cached FAISS IVF indexes and FAISS thread pinning, avoiding OpenMP oversubscription and cutting 155K-row neighbor search from 758us p50 flat FAISS to 154us p50 with IVF.
- Moved 9 live hot-path kernels into native Cython extensions for order-book maintenance, OBI, RTI, event aggregation, threshold storage, scalar math, and theo post-processing, cutting Kalshi book maintenance from 21.8ms to 195us p50 (112x) and RTI from 2.66ms to 323us p50 (8.3x).
- Built a compiled Cython/C++17 SPSC RTI event ring with cache-line-aligned atomic cursors, power-of-two masked slots, packed uint64 event records, per-venue/pair channels, single-producer enforcement, overflow-to-UNSYNCED recovery, and native ring-to-aggregator drains, benchmarked at 584ns enqueue and 703ns/event drain.
- Prevented bad-price quoting with fail-closed market-data and risk-admission gates that block stale, crossed, overflowed, low-venue, or untrusted feeds before quote generation; fused RTI/BRTI metrics benchmarked at 357us p50 / 432us p95 on 8 venues x 500 levels.
- Eliminated duplicate-send and crash-replay risk with a durable-before-wire order layer that journals intent before exchange dispatch, derives deterministic client IDs for idempotent retries, applies final pre-send risk gates, and persists ack/failure state for reconciliation.

EXPERIENCE

Software Engineering Intern

February 2026 – June 2026

RESET Standard

Shanghai, China

- Doubled the platform's supported environmental data categories (from 2 to 4) by independently implementing end-to-end water and waste data integration, developing Rails backend ingestion pipelines, PostgreSQL schemas, and client-facing React/Vite/Tailwind dashboards.
- Engineered asynchronous data pipelines with RabbitMQ, Redis, and Sidekiq to process long-running ingestion jobs in the background, decoupling external API calls from user-facing requests.
- Implemented a key client-requested feature by redesigning project-device mapping workflows, refactoring legacy forms to streamline the onboarding of environmental monitoring devices.

Technical Director

December 2025 – Present

Ezesports

New York, NY

- Migrated an esports league platform from Google Sites to Next.js with a Supabase backend, replacing manual content updates with live scoring and automated season management.
- Designed a PostgreSQL schema on Supabase using Drizzle ORM, supporting 500+ matches across 20+ teams, and built RESTful endpoints for player statistics, standings, and historical data.
- Cut page load time by 75% (from 3.2s to 0.8s) by implementing edge caching on Cloudflare, route-based code splitting, and asset optimization for the Next.js frontend.

TECHNICAL SKILLS

Languages: Rust, Cython, Python, TypeScript, JavaScript, Java, Ruby, SQL, HTML/CSS

Frameworks & Libraries: Next.js, React, FastAPI, Vite, Tailwind CSS, shadcn/ui, Ruby on Rails, Drizzle ORM

Databases & Infrastructure: PostgreSQL, Supabase, Redis, RabbitMQ, Sidekiq, Docker, Git, Cloudflare